

MEMORANDUM

Making Kestrel Festival Social-Listening Alerts More Trustworthy

To Marcus Vieira, Head of Social
From Sai Pranathi

Date 7 September 2026
Subject Kestrel social-listening recommendation

Executive recommendation

The command centre should treat social listening as an early-warning layer, not a standalone incident detector. The strongest design is to filter for genuine Kestrel/event relevance and human traffic, score sentiment, compare each market with its own pre-event baseline, and escalate only when negative movement is sustained.

1,206,715 posts analysed	97.0% automation precision	94.2% automation recall	1 / 4 negative incidents detected
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What was built

The final dataset contains 1.206M posts from 75,000 authors. It includes mixed timestamp formats, missing fields, late arrivals, unmatched author IDs and engagement inconsistencies. The pipeline audits these issues before analysis rather than assuming the feed is clean.

Of 325,831 posts mentioning Kestrel, 119,675 were classified as relevant, 120,365 ambiguous and 85,791 irrelevant. That means 206,156 Kestrel mentions (63.3%) were not admitted to the confirmed-relevance VSS set.

Automated traffic was handled with a transparent content-based detector. Against the private ground truth, 85,344 posts were flagged versus 87,868 true automated posts, giving 97.0% precision, 94.2% recall and a 0.632% error rate. Simple activity/repetition and coordination rules were tested and rejected because their false-positive rates were too high.

Sentiment uses a lightweight rule-based score with support for English, selected Hindi/Devanagari and Hinglish patterns, Portuguese/German terms, emoji, negation and selected sarcasm. Hindi negation remains a known weakness, so sentiment is treated as evidence rather than ground truth.

Decision rule: Verified Sentiment Shift (VSS)

VSS = current 15-minute net sentiment – same-market 14-day pre-event baseline, where net sentiment is % positive – % negative. Only human, confirmed-relevant posts with known market are used for VSS.

Volume gate	Shift gate	Escalation
< 220 qualifying posts → NO SIGNAL	$VSS \leq -14 \rightarrow$ breach	3 consecutive 15-min breaches → ESCALATE

A missing 15-minute window breaks continuity; one breach alone is not an escalation.

Evaluation: useful, but intentionally conservative

Across 280 market windows, the pipeline produced 35 breach windows and 20 escalation windows. In the six planted incidents, it detected the sponsor-related negative incident (30 minutes) and did not escalate either positive decoy. Overall detection was 1/6 incidents, or 1/4 negative incidents, with 0/2 positive decoys escalated.

The queue-crush, sound-failure and payment-outage incidents were missed. The evaluation does not prove a single root cause for those misses; plausible structural failure modes are low social visibility, short duration relative to the three-window rule, neutral operational language, and missing market metadata.

What the command centre should watch

- Sustained negative VSS: $VSS \leq -14$ for three consecutive 15-minute windows after the 220-post gate.
- Market concentration: negative movement concentrated in a known market rather than spread randomly across the feed.
- Human, relevant traffic: signals supported by confirmed Kestrel/event discussion after automated amplification and unrelated mentions are removed.
- Context: use the alert to trigger investigation alongside operational telemetry, not as proof of root cause.

What the command centre should ignore or down-rank

- Raw post-volume spikes without a corresponding negative sentiment shift.
- Generic or unrelated uses of “Kestrel”, including ambiguous mentions.
- Automated amplification, promotional/spam language and repeated low-information activity.
- A single VSS breach, a window below the 220-post gate, or a positive sentiment spike.
- Repeated text or coordinated-looking activity without independent evidence of an incident.

What the system cannot reliably see

- Low-volume incidents: events may fall below the 220-post gate.
- Short-lived incidents: the three-window rule deliberately trades recall for confidence.
- Operational issues with neutral language: people may report a failure without expressing strongly negative sentiment.
- Missing geography: market-level VSS cannot use posts without known market.
- Nuanced language: sarcasm and Hindi negation are weaker areas of the current sentiment rules.

Recommendation for production

Keep VSS as the conservative social-sentiment layer, but pair it with operational signals such as queue, sound and payment telemetry. For incidents that need faster detection, add a separate short-window operational detector rather than weakening the VSS thresholds. This preserves trust in the alert while covering failure modes that social sentiment alone cannot reliably observe.

Bottom line: Better monitoring is not about generating more alerts. It is about filtering aggressively, comparing against the right baseline, and escalating only when the evidence persists.